Stepan
Personal Care

Advanced Color Protection Systems

Stepan solutions for vibrant hair

Do not edit
How to change the design



What is the best way to achieve colour protection in shampoos?

The Slido app must be installed on every computer you're presenting from

slido

Stepan Personal Care

1. Surfactant screening

Choosing the right anionics

Anionic surfactants (and particularly sulfates) are the main ingredients responsible for color fading in shampoos: let's go for the most respectful ones!

Amphoterics matter too

Secondary surfactants play a great role in enhancing foam & mildness performances of a shampoo, but they also help in color protection

STEPAN-MILD® LSB/MB

Stepan Personal Care

Mildness and foam : ideal for color care

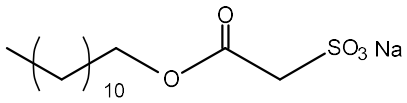
Identity Card

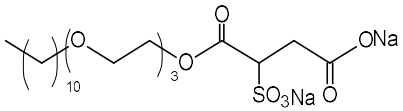
INCI Name

Sodium Lauryl Sulfoacetate and Disodium Laureth Sulfosuccinate

Chemical Structure

bci = 64





Typical Properties

Active (%)	25 %
Appearance (25°C)	Clear liquid
Preservative	Sodium Benzoate

Main benefits

Luxury foam

Color Care

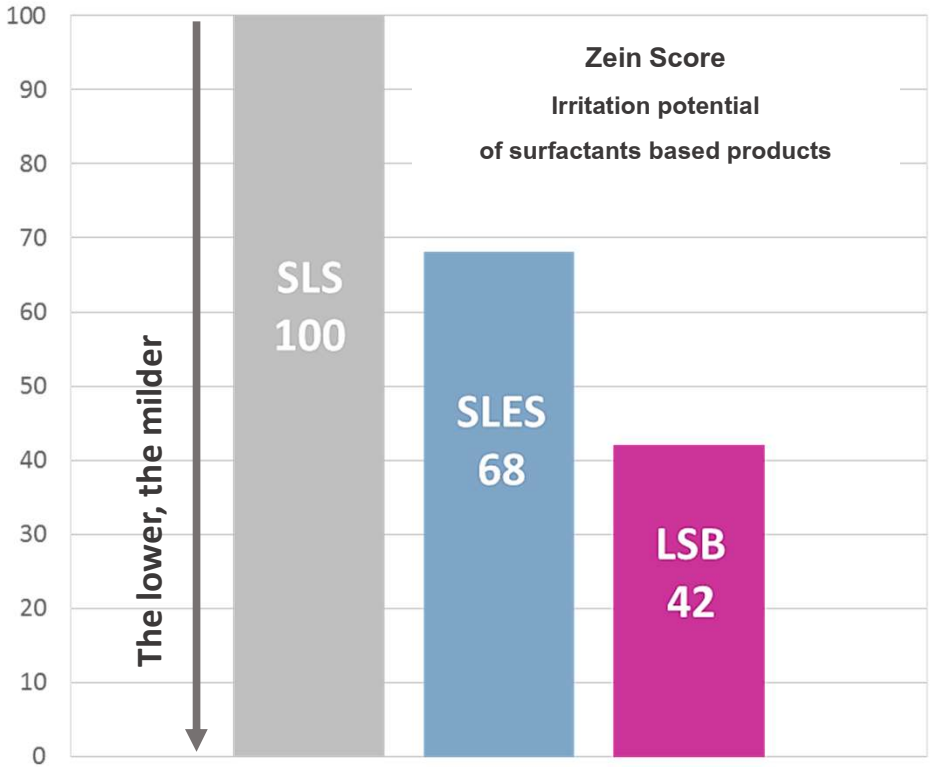
Mildness

STEPAN-MILD® LSB/MB

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Mildness

Luxury foam



Morning Caresse – Facial Cleanser



Ingredients	% active matter	%w/w
DI Water		Qs. 100
AMPHOSOL® DM/MB Coco betaine	3.9	13
STEPAN-MILD® LSB/MB Sodium Lauryl Sulfoacetate and Disodium Laureth Sulfosuccinate	1.6	6.4
STEPAN-MILD® GCC/MB Glyceryl Caprylate/Caprate	0.5	0.5
Perfume		0.4
Potassium sorbate, Sodium benzoate		1.2
pH adjuster		Qs. to pH = 5,5

Stability :

1 month @45°C

LATHANOL® LAL Coarse/MB

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Dense and creamy foam

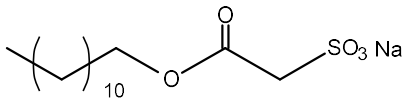
Identity Card

INCI Name

Sodium Lauryl Sulfoacetate

Chemical Structure

bci = 87



Typical Properties

Active (%)	65 % min
Appearance (25°C)	White Coarse

Main benefits

Creamy foam

Secondary or additive

Mildness

BIO-TERGE® AS-40 HASB

Stepan Personal Care

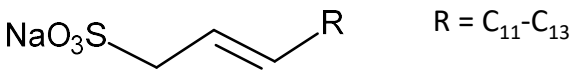
Solution for cost optimised sulfate free products

Identity Card

INCI Name

Sodium C14-C16 Olefin sulfonate

Chemical Structure



$R = C_{11}$-C_{13}

Typical Properties

Active (%)	37-39 %
Appearance (25°C)	Clear Liquid
Palm/palm kernel Oil free	

Main benefits

High foam

SLES Substitute

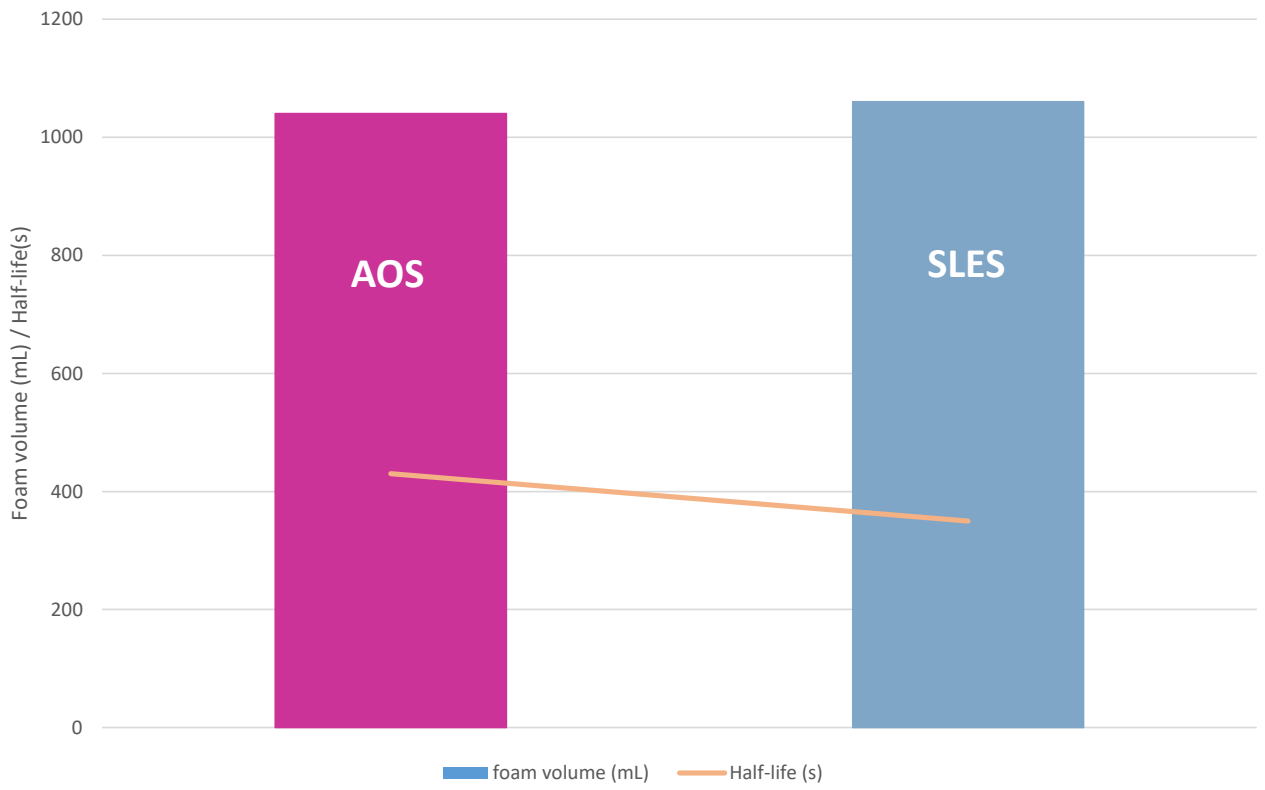
pH stability

BIO-TERGE® AS-40 HASB

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High Foam

Similar foaming behaviour compared to SLES (Shaker method)

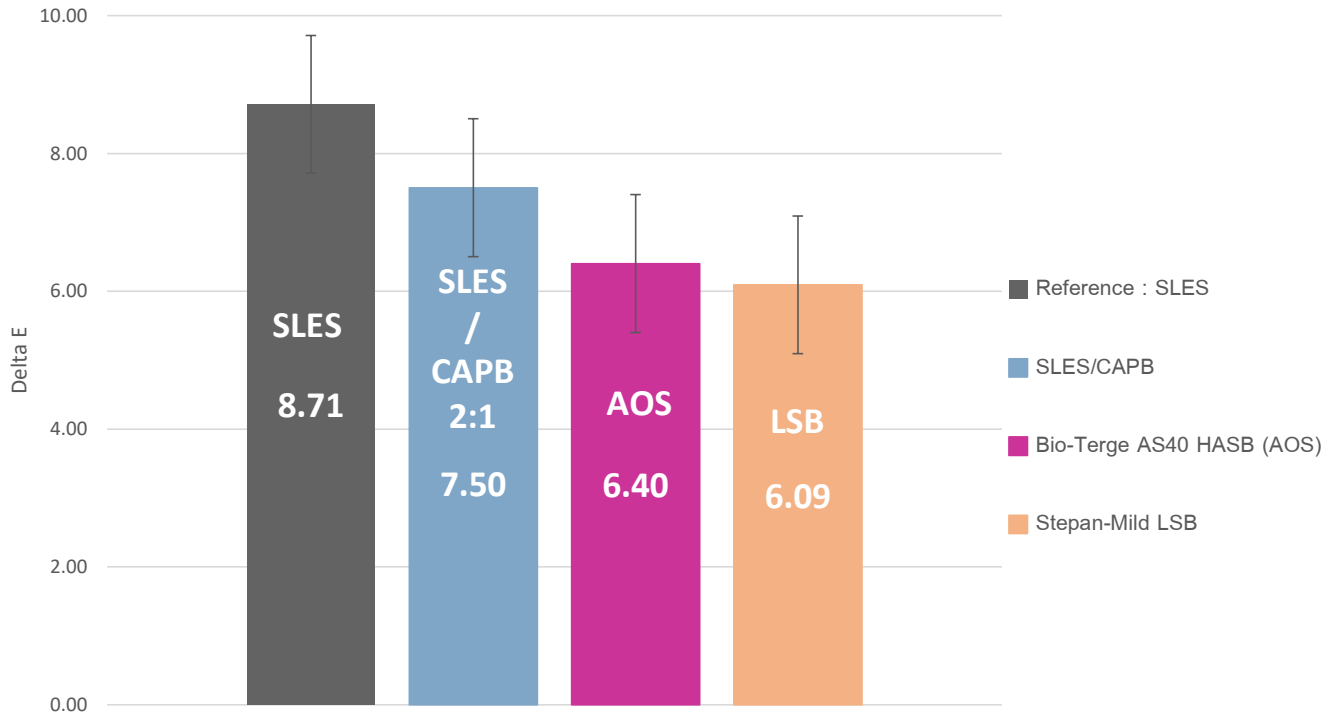


Test at 0,4% surfactant active matter, DI water

Color Care

Primary Surfactants

Color Fading Evaluation – L*a*b* measurement



Both BIO-TERGE® AS-40 HASB & STEPAN-MILD® LSB/MB show the greatest color respect, even compared to a I^{ary}/II^{ary} system

Method

European bleached hair wefts dyed in red

(100% Ultra Color 665 Ultra Rouge from Garnier)

2 ultra-sounds cycles

10 min, 30°C

3% actives dilution

500mL / 3 wefts

Stepan Personal Care

1. Surfactant screening

Choosing the right anionics

Anionic surfactants (and particularly sulfates) are the main ingredients responsible for color fading in shampoos: let's go for the most respectful ones!

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Secondary surfactants play a great role in enhancing foam & mildness performances of a shampoo, but they also help in color protection

AMPHOSOL® DM/MB

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The choice amphoteric - CAPB substitute

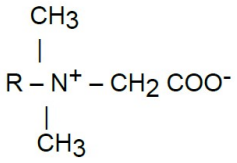
Identity Card

INCI Name

Coco Betaine

Chemical Structure

bci = 76



R = C_{12}/C_{14} maj. (typ. value 95%)

Typical Properties

Active (%)	30 %
Appearance (25°C)	Clear Liquid
No DMAPA residue	

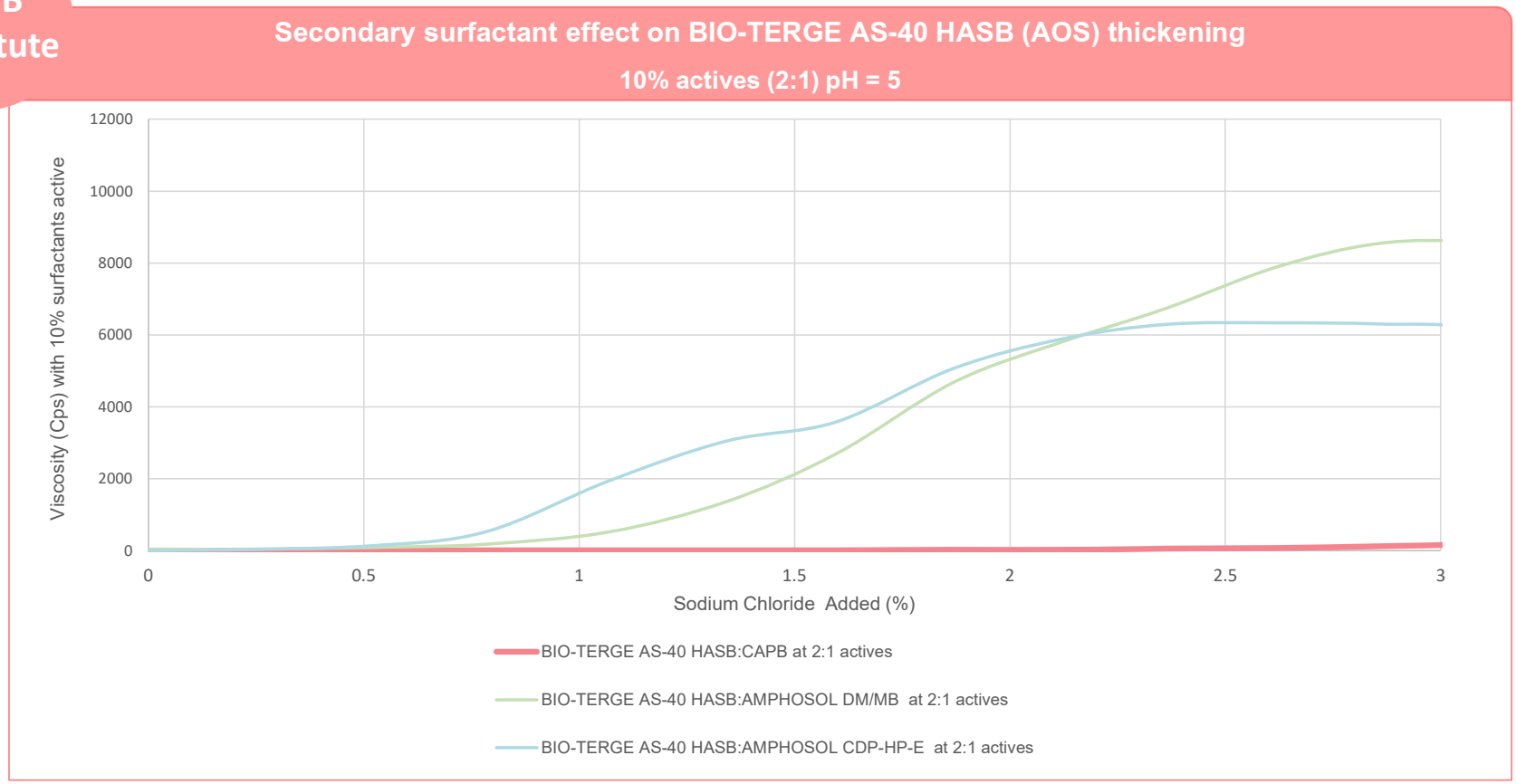
Main benefits

- Mildness
- Color Care
- Viscosity building

Viscosity building

AMPHOSOL DM/MB

CAPB substitute



Formulation

Stepan Personal Care

Sulfate free

Foam & Viscosity

Lotus Night– Sulfate Free Body Wash



Crystal clear @ RT

Stability : 1 month @ 5°C and 45°C

Viscosity (LV2,v60, 20°C) : about 3000cP

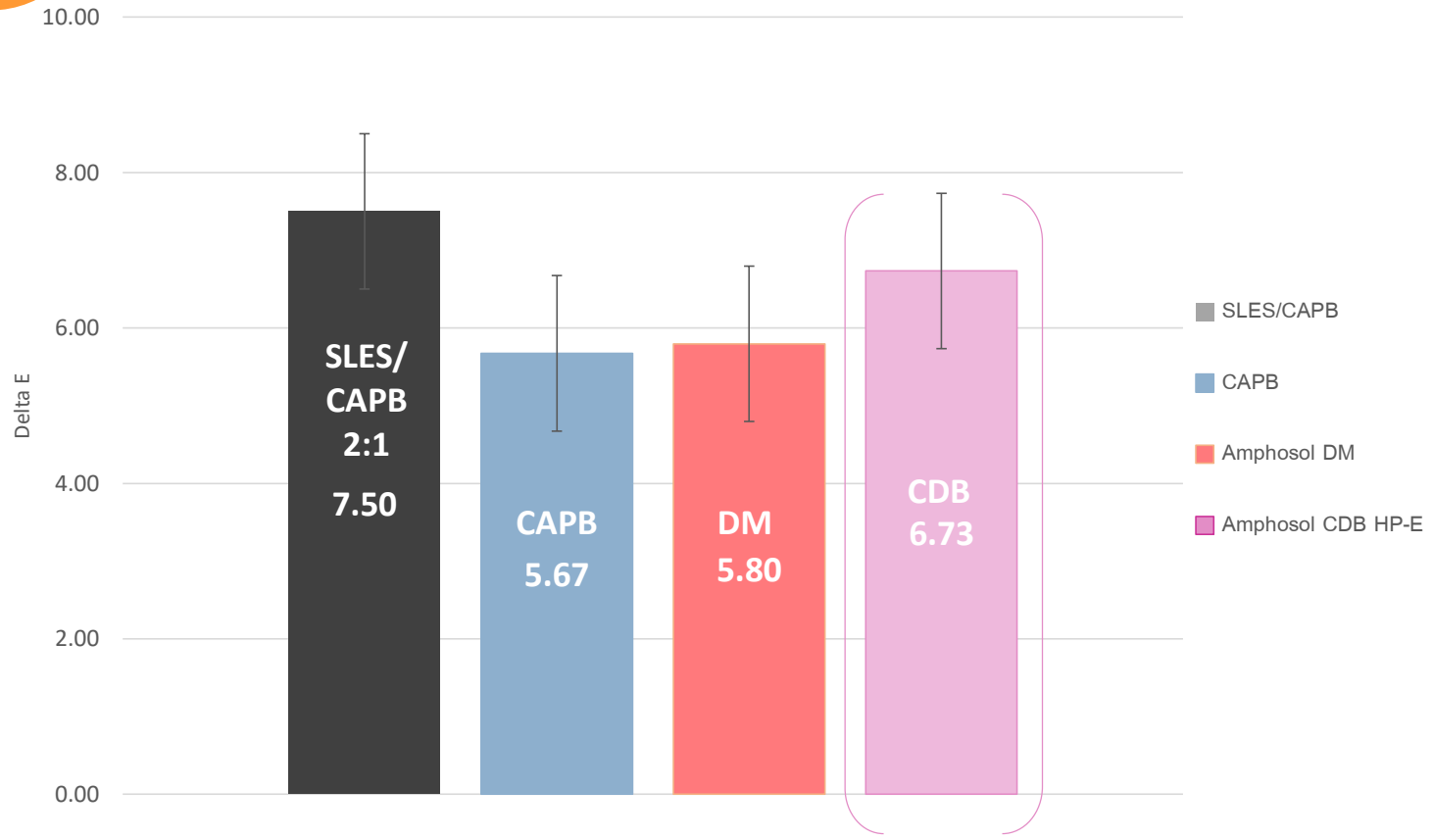
Ingredients	% active	%w/w
DI Water		Qs. 100
BIO-TERGE® AS-40 HASB Sodium C14-16 Olefin Sulfonate	8	20
AMPHOSOL® DM/MB Cocobetaine	4,5	15
STEPAN-MILD® L3-G/MB Lauryl Lactyl Lactate	0,5	0,5
Perfume		Qs.
Potassium sorbate, Sodium benzoate		1,2
pH adjuster		Qs. to pH < 5,5

Color Care

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Secondary Surfactants

Color Fading Evaluation – L*a*b* measurement



Method

European bleached hair wefts dyed in red

(100% Ultra Color 665 Ultra Rouge from Garnier)

2 ultra-sounds cycles

10 min, 30°C

3% actives dilution

500mL / 3 wefts

AMMONYX® LO & AMMONYX® CSO

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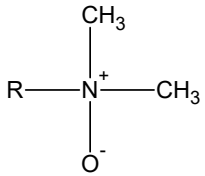
Identity Card

INCI Name

Lauramine Oxide / C12-18 Amine Oxide

Chemical Structure

bci = 86 / 87



AMMONYX® LO : $R_{typ.}$ = C_{12}/C_{14} (70/30)

AMMONYX® CSO : $R_{typ.}$ = C_{12}/C_{14} (>60%) and C_{16}/C_{18}

Typical Properties

Active (%)	30 %
Appearance (25°C)	Clear Liquid

Main benefits

High foaming power

Color Care

Cationic when ph < 6,5

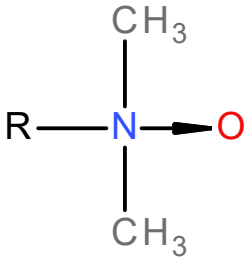
AMMONYX® LO & AMMONYX® CSO

Stepan Personal Care

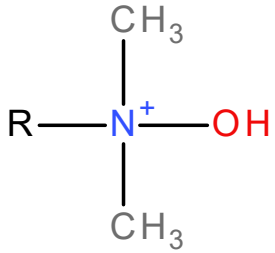
Benefits from skin to hair



- Strong dipolar moment between O and N
- Reduces irritancy of primary surfactants
- Good skin compatibility



pH > 6,5



pH < 6,5

- Cationic charge
- Anti-static properties
- Compatibility with quats
- Suitable for colored shampoo

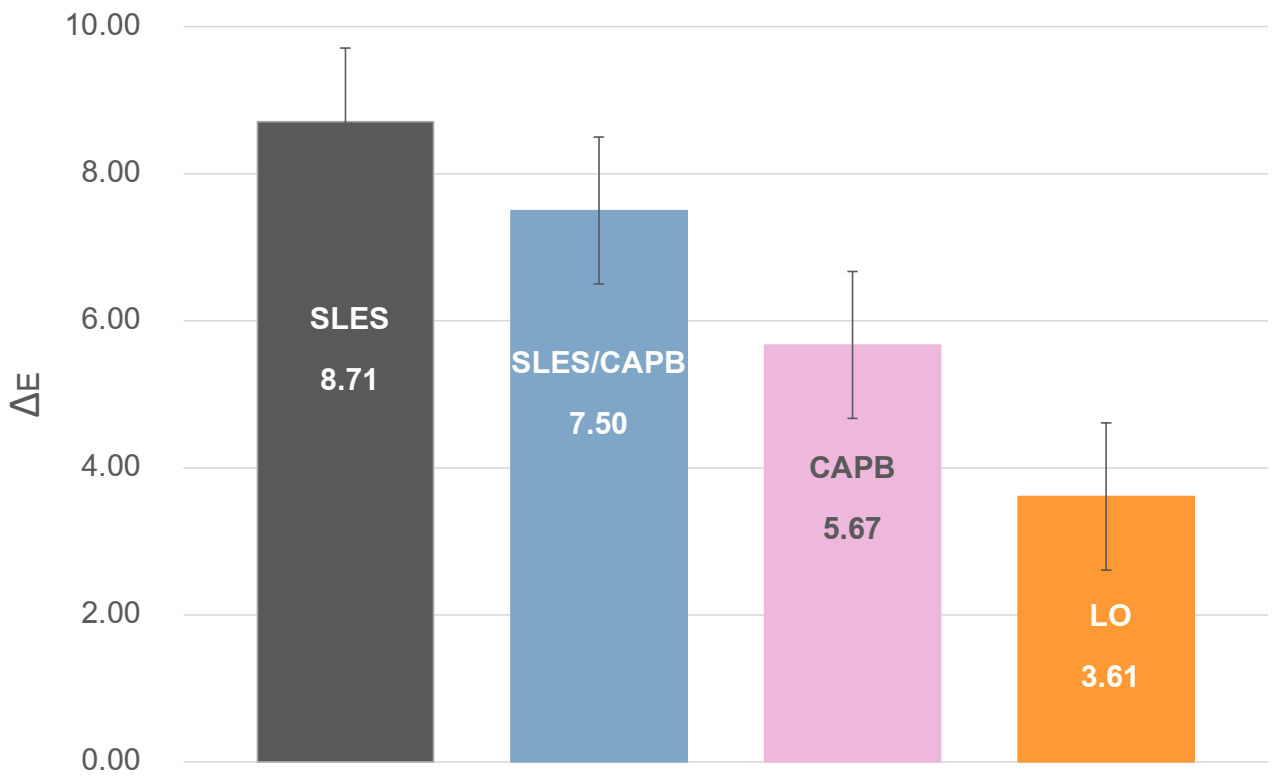


Color Care

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AMMONYX LO

Color Fading Evaluation – L*a*b* measurement



Method

European bleached hair wefts dyed in red

(100% Ultra Color 665 Ultra Rouge from Garnier)

2 ultra-sounds cycles

10 min, 30°C

3% actives dilution

500mL / 3 wefts

Stepan Personal Care

2. Advanced color protection systems

High performance shampoo

Optimized selection of Stepan ingredients that combine high color protection level with delivering superior sensory benefits

Getting out of the box

Next generation systems for even improved performance: anionic-free shampoo & cleansing conditioner

Formulation inspiration

Stepan Personal Care

Sulfate-Free Color Care Shampoo

Formulation No 1605

Hair that has been dyed requires special care. This sulfate-free shampoo is specially formulated for color treated hair. It effectively cleanses without causing damage, thanks to the gentle surfactants STEPAN-MILD LSB/MB, BIO-TERGE AS-40 HASB, and AMPHOSOL DM/MB. Your hair will feel supple and soft due to RSPO Mass Balance, natural and COSMOS certified emollients: STEPAN-MILD L3-G/MB and STEPAN-MILD GCC/MB. Additionally, these ingredients enhance foam volume and creaminess, providing a luxurious washing experience. Your hair will remain luminous and shiny!.

Ingredients	% by weight
Deionized Water	q.s. to 100.0
STEPAN-MILD LSB/MB *Disodium Laureth Sulfosuccinate (and) Sodium Lauryl Sulfoacetate*	**32.0**
BIO-TERGE AS-40 HASB *Sodium C14-16 Olefin Sulfonate*	**16.2**
AMPHOSOL DM/MB *Coco Betaine*	**6.0**
STEPAN-MILD L3-G/MB *Lauryl Lactyl Lactate*	**1.0**
STEPAN-MILD GCC/MB *Glyceryl Caprylate/Caprate*	**1.0**
Citric Acid	q.s
Fragrance, preservative, dye	q.s

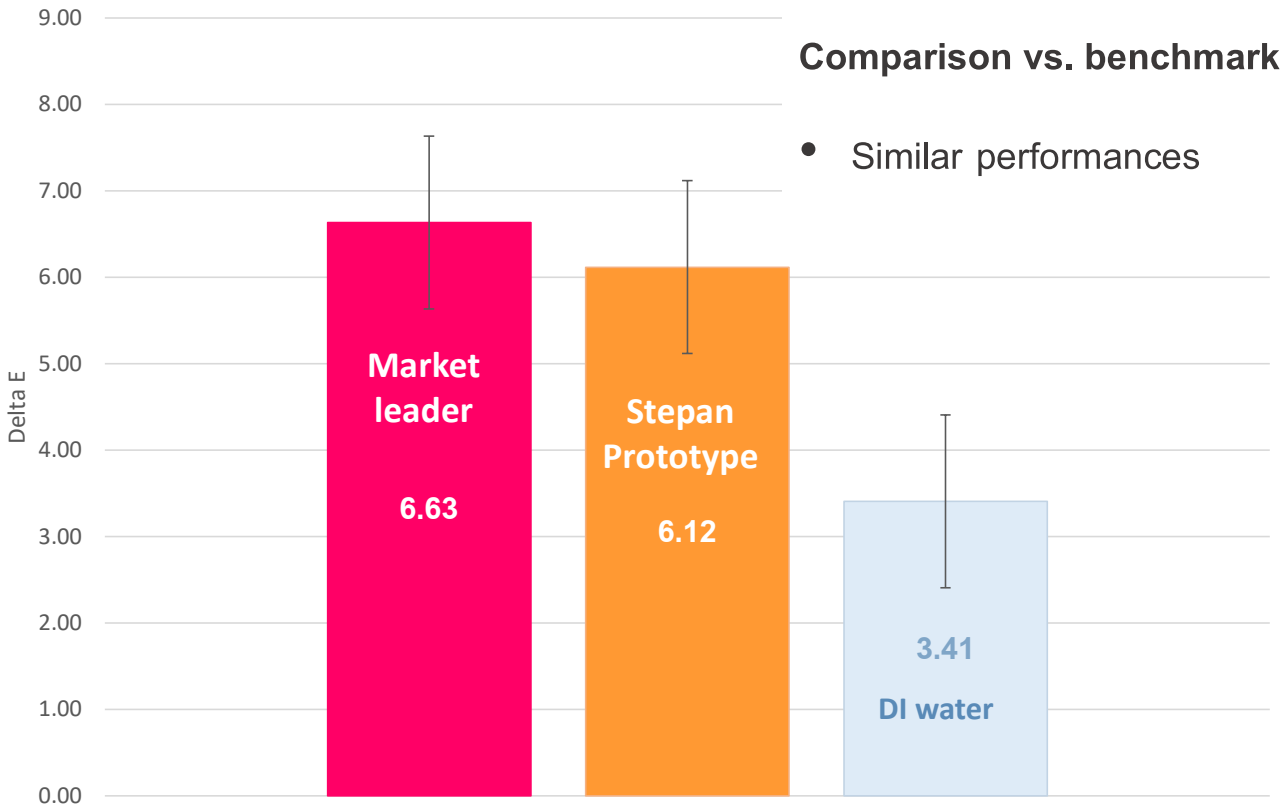


Color Care

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Stepan prototype

Color Fading Evaluation – L;a;b measurement



Comparison vs. benchmark

- Similar performances

Method

European bleached hair wefts dyed in red

(100% Ultra Color 665 Ultra Rouge from Garnier)

2 ultra-sounds cycles

10 min, 30°C

3% actives dilution

500mL / 3 wefts

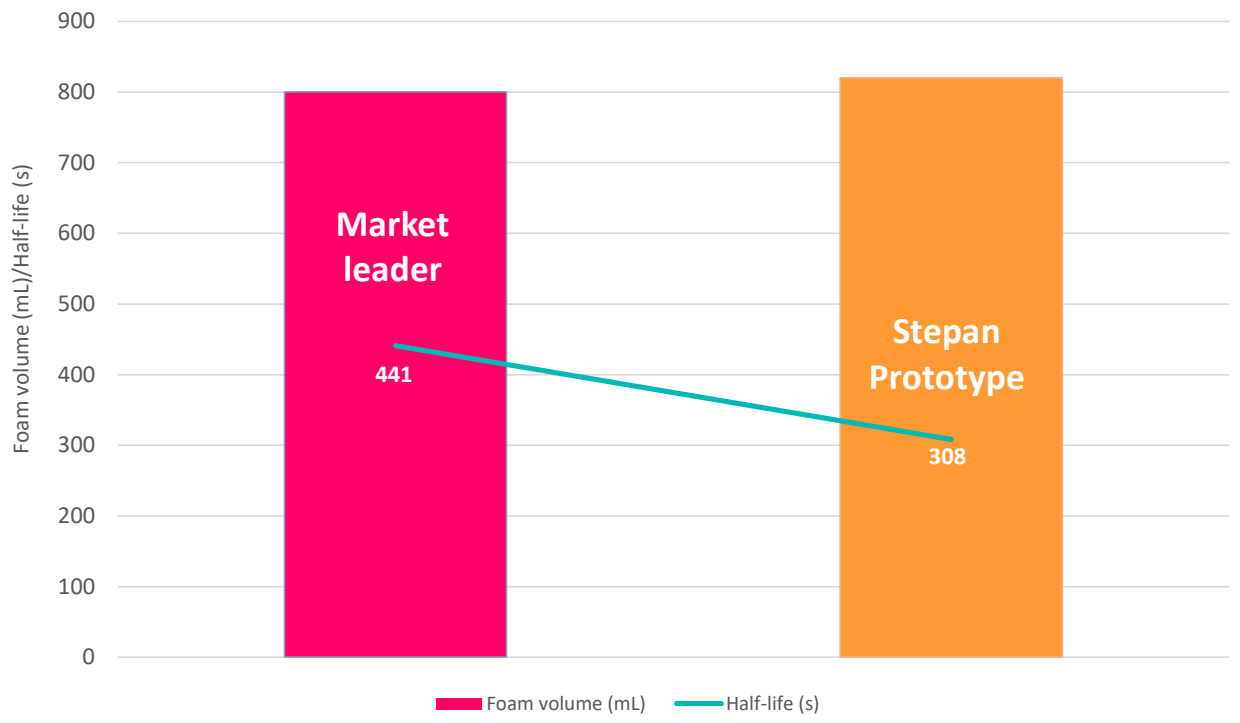
Foam evaluation

Stepan Personal Care

Stepan Prototype



Foam volume and stability (Shaker method)



Comparison vs. Benchmark

- Similar foam volume
- Slightly lower foam stability

2. Advanced color protection systems

High performance shampoo

Optimized selection of Stepan ingredients that combine high color protection level with delivering superior sensory benefits

Getting out of the box

Next generation systems for even improved performance: anionic-free base & cleansing conditioner

Color Care

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Very Mild Anionic-free chassis

Ingredients	% active	%w/w
DI Water		Qs. 100
AMMONYX® CSO C12-18 Amine Oxide	9	30
AMPHOSOL® DM/MB Cocobetaine	3	10
STEPAN-MILD® L3-G/MB Lauryl Lactyl Lactate	1,2	1,2
STEPAN-MILD® GCC/MB Glyceryl Caprylate/Caprate	1,2	1,2
Preservative, addtitives		Qs. 100
pH adjuster		Qs. to pH < 5,5



Color Care

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Cleansing conditioner with color care benefits

Ingredients	% active	%w/w
DI Water		Qs. 100
AMMONYX® LO Lauramine Oxide	6	20
Stepan-Mild® GCC/MB Glyceryl Caprylate Caprate	6	6
STEPAN-MILD® L3-G/MB Lauryl Lactyl Lactate	1	1
STEPANQUAT® GA 90/MB Dipalmitoylethyl Hydroxyethylammonium Methosulfate	2	2,22
Decyl Glucoside	2	4
Cocamide MIPA	0,8	0,8
Sunflower oil	4	4
Jojoba Oil	1	1
Perfume, preservative		qs
Lactic acid		Qs. to pH < 4,5

Structured liquids technology

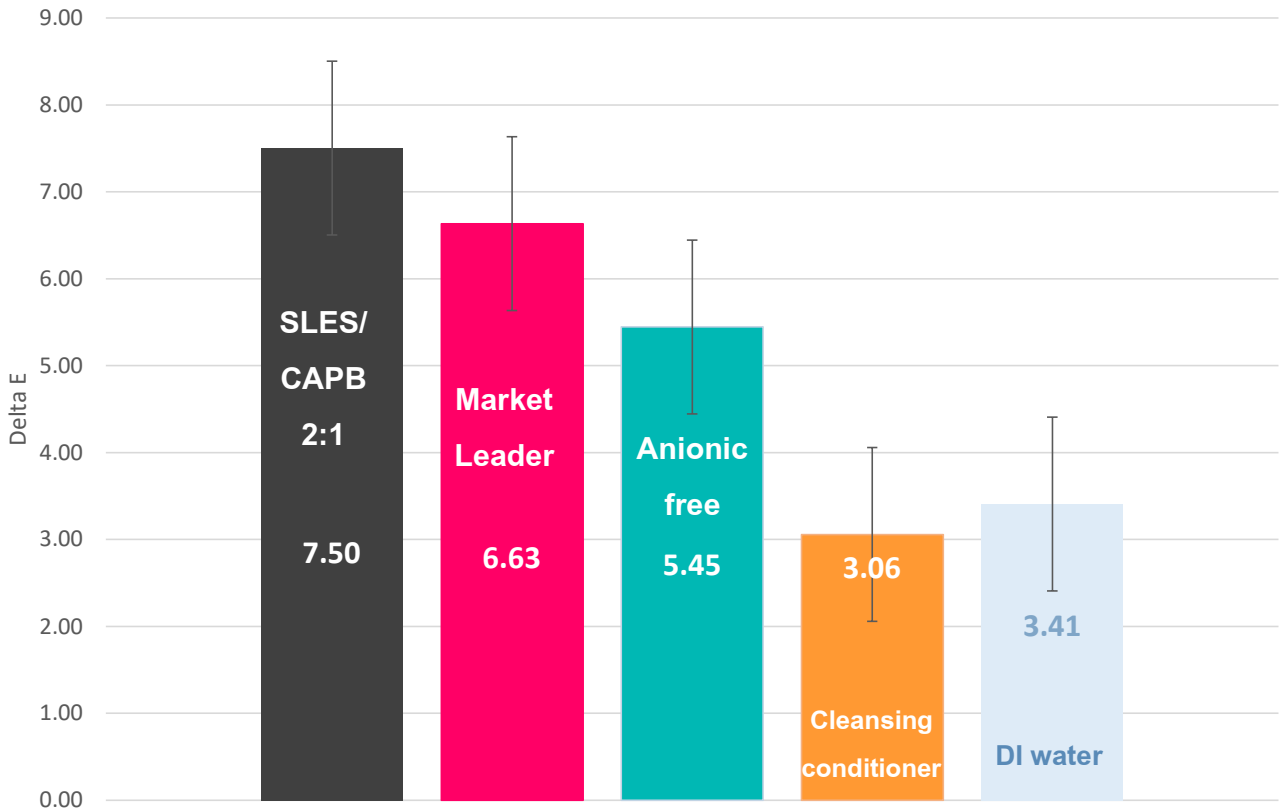


Color Care

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Next-gen systems

Color Fading Evaluation – L*a*b* measurement



(Stepan shampoo prototype: ΔE = 6,12)

Method

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2 ultra-sounds cycles

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500mL / 3 wefts

Page withheld in this copy: it held personal contact details, not seminar content.